

Brain tumour classification from MRI using two-phase transfer learning

A ResNet18 fine-tuned in two stages classifies axial brain MRI slices into four categories at 97.00% test accuracy — and concentrates almost two-thirds of its residual error on the one boundary that is hardest for human readers too.

Architecture ResNet18, ImageNet pretrained **Dataset** BRISC2025 · 6,000 scans **Framework** PyTorch · torchvision · Flask

97.00%

Test accuracy — 970 / 1,000 held-out scans

0.9715

Macro-averaged F1

100%

Recall on *no tumour* — zero missed tumours

11.2M

Parameters · 8.4 M fine-tuned

THE PROBLEM

Tumour type must be identified before treatment can be planned, but reading MRI is slow, needs scarce expertise, and varies between observers — most of all between **glioma** and **meningioma**, which can look alike on a single axial slice.

We classify one slice into four classes:

Glioma Meningioma No tumour Pituitary

5,000 training images is far too few to train an 11-million-parameter network from scratch, so the whole method turns on **how** pretrained weights are adapted.

DATA

SPLIT	IMAGES	SOURCE
Train	4,000	80% of train dir
Validation	1,000	20%, stratified
Test	1,000	held out
Total	6,000	seed 42

PREPROCESSING

1 Resize to 224 × 224

The resolution ResNet18 was pretrained at.

2 Greyscale → 3 channels

Replicating the single channel lets the pretrained RGB filters apply unchanged.

3 Horizontal flip only

Anatomy is near-symmetric, so a flip is plausible. Rotation and jitter were excluded — fabricating non-physical scans teaches artefacts, not pathology.

4 ImageNet normalisation

Places inputs in the distribution the pretrained weights expect.

METHOD — WHY TWO PHASES

The new 4-class head starts random. Unfreezing everything at once would push large, noisy gradients back through the backbone and **destroy the pretrained features before they could be used**. Only **layer4** is ever unfrozen: early layers already hold generic edge and texture detectors, while the deepest block holds the class semantics that must move from natural images to MRI.

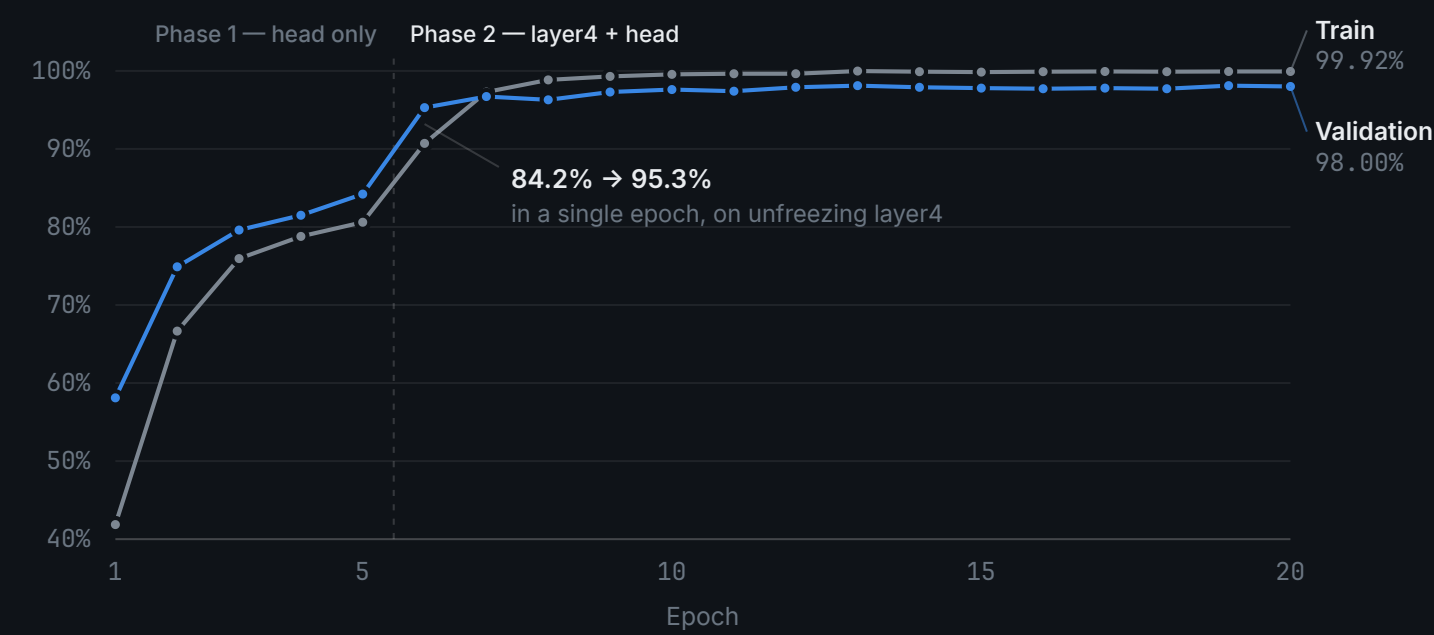
Phase 1 · head only

Epochs	5
Trainable	2,052
LR (Adam)	1e-4
Backbone	frozen
Val acc	84.2%

Phase 2 · layer4 + head

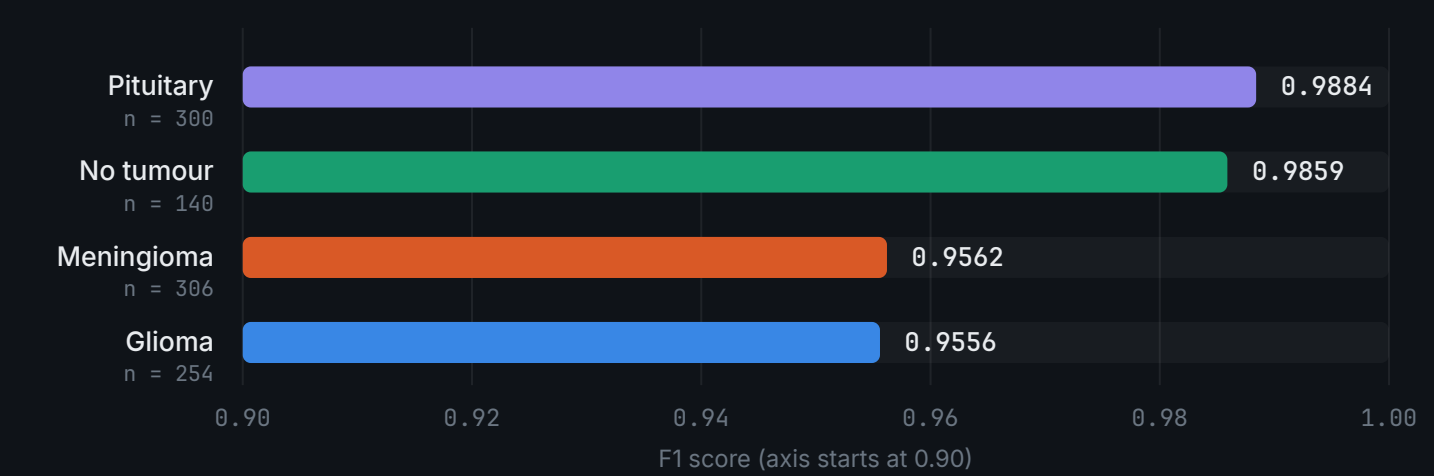
Epochs	15
Trainable	8,395,780
LR (Adam)	5e-5
Schedule	StepLR
Val acc	98.0%

TRAINING — VALIDATION ACCURACY ACROSS BOTH PHASES



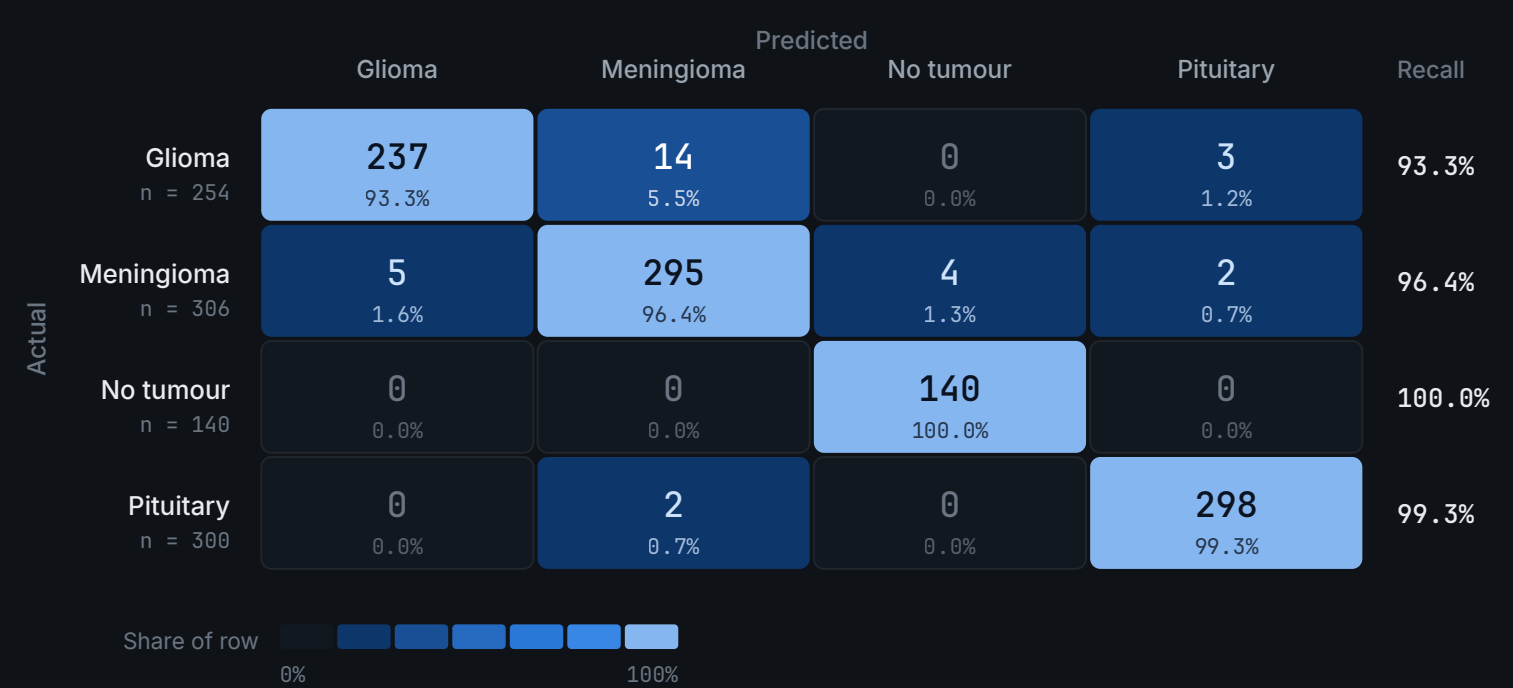
Phase 1 plateaus at **84.2%**: a frozen ImageNet backbone can only reach so far into a domain as distant as MRI. Unfreezing **layer4** lifts validation accuracy to **95.3% in a single epoch**, and past 97% by epoch 9. By epoch 13 it has converged: train sits at 99.9%+ while validation holds a narrow 97.7–98.1% band.

RESULTS — PER-CLASS F1 ON THE HELD-OUT TEST SET



Glioma and meningioma trail together — the first sign they are being confused with *each other* rather than failing independently. Macro and weighted F1 agree to within **0.002**, so the headline is not an artefact of class imbalance.

CONFUSION MATRIX — 30 ERRORS IN 1,000 SCANS



WHAT THE ERRORS SAY

63% of errors

Glioma ↔ meningioma is the one real weakness. 19 of 30 errors sit on this pair — also the hardest call for human readers on a single slice. The model fails where the task is genuinely ambiguous, not at random.

0 missed

Perfect recall on *no tumour*. No scan containing a tumour was called tumour-free. This class's four errors all run the safe way, so false negatives — the costliest error in screening — did not occur.

99.3% recall

Pituitary is nearly separable. Two errors in 300. These tumours sit at a distinctive skull-base location, giving the network a positional cue the other tumour types do not offer.

LIMITS & NEXT STEPS

Limits. One slice, not the volume. No localisation. A single dataset, so cross-scanner generalisation is unmeasured. Confidence is raw softmax and uncalibrated.

Next. Target the glioma/meningioma boundary with a focal loss or a second-stage binary head; add Grad-CAM; validate externally; aggregate across the slices of a study.